Zikai Xiong

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Academic Appointments

Northwestern University, Evanston, IL *July 2026 – present*
Assistant Professor
Department of Industrial Engineering and Management Sciences

Georgia Institute of Technology, Atlanta, GA *July 2025 – July 2026*
Postdoctoral Fellow
School of Industrial and Systems Engineering

Research Interests

Optimization theory and computation, particularly **large-scale optimization on AI-era computing hardware** and **optimization with humans and AI**, with applications in operations research, machine learning, and AI.

Education

Massachusetts Institute of Technology, Cambridge, MA *2020 – 2025*
Ph.D. in Operations Research
Thesis: *New Theory and New Practical Methods for Solving Large-Scale Linear and Conic Optimization* [link] [pdf]
- Second Place, 2025 George B. Dantzig Dissertation Award

Advisor: Prof. Robert M. Freund

Fudan University, Shanghai, China *2016 – 2020*
B.S. in Mathematics and Applied Mathematics

Publications (alphabetical authorship marked with *)

Papers Published or Under Review at Journals

J1 Zikai Xiong. High-Probability Polynomial-Time Complexity of Restarted PDHG for Linear Programming. [arXiv] [pdf]
Mathematics of Operations Research (forthcoming)

J2 Zikai Xiong. Accessible Complexity Bounds for Restarted PDHG on Linear Programs with a Unique Optimizer. [link] [pdf]
Mathematics of Operations Research, 2026
- Winner, 2025 INFORMS Optimization Society Student Paper Prize

J3 Zikai Xiong and Robert M. Freund. Computational Guarantees for Restarted PDHG for LP based on "Limiting Error Ratios" and LP Sharpness. [link] [pdf]
Mathematical Programming, 2026
- Honorable Mention, 2024 INFORMS Optimization Society Student Paper Prize

J4 *Katya Scheinberg and Zikai Xiong. Function-free Optimization via Comparison Oracles. [arXiv] [pdf]
Under revision at *Mathematical Programming*, *2026*

J5 Zikai Xiong and Robert M. Freund. Symmetry-dependence in Rounding of a Convex Body. [arXiv] [pdf]
Under review at *Mathematical Programming*, *2026*

J6 Zikai Xiong and Robert M. Freund. The Role of Level-Set Geometry on the Performance of PDHG for Conic Linear Optimization. [arXiv] [pdf], *2024*
A subsequent version, "Level-Set Geometry and the Theoretical Performance of PDHG for Conic Linear Optimization," is under review at *Mathematics of Operations Research*. [arXiv] [pdf]
- Finalist, 2024 George Nicholson Student Paper Competition
- Honorable Mention, 2024 MIT ORC Best Student Paper Competition

J7 *Dongdong Ge, Chengwenjian Wang, Zikai Xiong and Yinyu Ye. From an Interior Point to a Corner Point: Smart Crossover. [link] [pdf]
INFORMS Journal on Computing, 2025

J8 Zhenwei Lin, Zikai Xiong, Dongdong Ge and Yinyu Ye. A Practical GPU-Enhanced Matrix-Free Primal-Dual Method for Large-Scale Conic Programs. [arXiv] [pdf] [additional results] [software]
Under revision at *INFORMS Journal on Computing*, *2025*

J9 *Jiacheng Guo, Bo Jiang, Zikai Xiong and Nanxi Zhang. The Impact and Insights of Hidden City Ticketing: Analysis of a Choice-Based Network Revenue Management Model. [link]
Under revision for resubmission at *Manufacturing & Service Operations Management*, *2025*

J10 Zikai Xiong and Robert M. Freund. Using Taylor-Approximated Gradients to Improve the Frank-Wolfe Method for Empirical Risk Minimization. [link] [pdf] [additional results]
SIAM Journal on Optimization, 2024

J11 Zikai Xiong, Yimin Wei, Renjie Xu and Yanwei Xu. Low-rank Traffic Matrix Completion with Marginal Information. [link] [pdf]
Journal of Computational and Applied Mathematics, 2022

Papers Published at Conferences

C1 Zikai Xiong, Niccolò Dalmasso, Shubham Sharma, Freddy Lecue, Daniele Magazzeni, Vamsi K. Potluru, Tucker Balch and Manuela Veloso. Fair Wasserstein Coresets. [link] [pdf]
Preliminary version appeared in NeurIPS 2023 Workshop SyntheticData4ML (Oral Presentation)
Neural Information Processing Systems (NeurIPS), 2024

C2 Zikai Xiong, Niccolò Dalmasso, Alan Mishler, Vamsi Potluru, Tucker Balch and Manuela Veloso. FairWASP: Fast and Optimal Fair Wasserstein Pre-processing. [link] [pdf]
AAAI Conference on Artificial Intelligence (AAAI), 2024

C3 Zhengqi Gao, Fan-Keng Sun, Mingran Yang, Sucheng Ren, Zikai Xiong, et al. Learning from Multiple Annotator Noisy Labels via Sample-wise Label Fusion. [link] [pdf]
European Conference on Computer Vision (ECCV), 2022

C4 *Dongdong Ge, Haoyue Wang, Zikai Xiong and Yinyu Ye. Interior-Point Methods Strike Back: Solving the Wasserstein Barycenter Problem. [link] [pdf]
Neural Information Processing Systems (NeurIPS), 2019

Technical Reports

T1 Zhenwei Lin, Zikai Xiong, Dongdong Ge and Yinyu Ye. A Technical Note on the Implementation and Use of PDCS [arXiv] [pdf], *2026*

T2 Zikai Xiong and Robert M. Freund. On the Relation Between LP Sharpness and Limiting Error Ratio and Complexity Implications for Restarted PDHG. [arXiv] [pdf], *2023*

Patent Applications

P1 Niccolò Dalmasso, Zikai Xiong, Vamsi Krishna Potluru, Tucker Richard Balch and Manuela Veloso. Method and System for Generating Fair Synthetic Representative Data via Optimal Transport. [link]
U.S. Patent Application Publication No. US 2025/0181999 A1, pending, 2025

P2 Niccolò Dalmasso, Zikai Xiong, Alan Mishler, Vamsi Krishna Potluru, Tucker Richard Balch and Manuela Veloso. Method and System for Pre-processing Data for Algorithmic Fairness via Optimal Transport. [link] *U.S. Patent Application Publication No. US 2025/0173389 A1, pending, 2025*

Software

- **PDCS**: a GPU-enhanced matrix-free primal-dual solver for large-scale conic programs, with JuMP and CVXPY interfaces. [link]
- **Smart Crossover**: a Python package that finds optimal basic solutions from interior-point solutions. [link]
- **PDLP Complexity Bound**: a Julia toolkit that computes the accessible iteration bounds of restarted PDHG for LP. [link]
- **MAAIPM**: a MATLAB solver for exactly solving Wasserstein barycenter problems using interior-point methods. [link]

Awards and Honors

- NSF Travel Award for Young Researchers, Foundations of Computational Mathematics (FoCM) *2026*
- **Second Place, George B. Dantzig Dissertation Award** *2025*
- **Winner, INFORMS Optimization Society Student Paper Prize** *2025*
- **Finalist, George Nicholson Student Paper Competition** *2024*
- **Honorable Mention, INFORMS Optimization Society Student Paper Prize** *2024*
- **Honorable Mention, MIT ORC Best Student Paper Competition** *2024*
- INFORMS Optimization Society Conference Travel Award *2024*
- MIT Graduate Student Council Travel Grant *2022*
- First Prize, MIT ORC Common Experience Deep Learning Challenge *2021*
- SIAM Travel Award *2021*
- Star Graduate Award, Fudan University *2020*
- Outstanding Graduate of Shanghai City *2020*
- Top Outstanding Undergraduate Student Award, Fudan University *2019*

Work Experience

Georgia Institute of Technology, Postdoctoral Fellow, Atlanta, GA *July 2025 - July 2026*
Mentor: Katya Scheinberg
- Optimization with human and AI feedback; function-free optimization via comparison oracles.

MIT Sloan School of Management, Research Assistant, Cambridge, MA *2020 - 2025*
Supervisor: Robert M. Freund
- Large-scale optimization on AI-era computing hardware; understanding and improving first-order methods for linear and conic optimization.

J.P. Morgan AI Research, Research Associate Intern, New York, NY *Summer 2023*
Managers: Niccolò Dalmasso, Vamsi Potluru and Manuela Veloso
- Algorithmic fairness in machine learning and large-scale optimization; led to papers at NeurIPS 2024 and AAAI 2024 and two U.S. patent applications.

Shanghai University of Finance and Economics (SUFE), Research Assistant, Shanghai, China *2019 - 2020*
Supervisors: Yinyu Ye (Stanford), Dongdong Ge
- Optimization for operations research and machine learning.
- Developed and improved algorithms for the commercial solver COPT.

Teaching

Northwestern University (Instructor)

Nonlinear Optimization for Decision-Making (IEMS 314, undergraduate) *Spring 2027 (scheduled)*

Mathematical Optimization II (IEMS 450-2, graduate) *Winter 2027 (scheduled)*

MIT

Computing in Optimization and Statistics (15.S60) *Winter 2024*

- Instructor for the advanced optimization sessions of a computing course during MIT January Independent Activities Period. The duties include designing lectures, creating teaching materials, and giving lectures.

Introduction to Mathematical Programming (15.081) *Fall 2022 and Fall 2023*

- Head TA for MIT's doctoral introductory optimization course (for ORC and other MIT PhD programs).
- Teaching score: 6.8/7 (Fall 2023) and 6.6/7 (Fall 2022).
- Voted as the "Best TA" in the ORC Superlatives events (in 2023 and 2024).

The Analytics Edge (15.071) *Spring 2022*

- TA for MBA course in business analytics at Sloan School of Management.
- Teaching score: 6.4/7.

Online Course in Data Science and Business Analytics *Summer 2022*

- Designed course materials for a new online course in data science and business analytics.
- Supervisor: Robert M. Freund

Guest Lectures

- Research talk on function-free optimization for undergraduates, IDEAL Get Ready for Research Workshop
  University of Illinois Chicago, Chicago, IL *October 2026*
- Large-Scale Linear Programming Using the Primal-Dual Hybrid Gradient Method
  MIT, Cambridge, MA *November 2025*

Advising

PhD Students, Northwestern University

- Hongpei Li *2026 - present*
- Jinrui Huang *2026 - present*

Students Mentored Before Northwestern

- Qipei Chen, visiting undergraduate, University of California, Berkeley
- Zhenwei Lin, PhD student, Shanghai University of Finance and Economics *(later a postdoctoral researcher at Purdue University)*
- Qiushi Han, undergraduate, University of Illinois Urbana-Champaign *(later joined the PhD program at MIT)*
- Chenghan Xie, undergraduate, Fudan University *(later joined the PhD program at Stanford)*
- Jiacheng Guo, undergraduate, Fudan University *(later joined the PhD program at Princeton)*
- Chengwenjian Wang, undergraduate, Fudan University *(later joined the Master program at ETH Zürich and the PhD program at the University of Minnesota Twin Cities)*

Talks

Function-Free Optimization via Comparison Oracles

- INFORMS Annual Meeting, San Francisco, CA *November 2026*
- Research Institute for Interdisciplinary Sciences, Shanghai University of Finance and Economics, Shanghai, China *September 2026*
- Modeling and Optimization: Theory and Applications (MOPTA), Lehigh University, PA *August 2026*
- Foundations of Computational Mathematics (poster), University of Vienna, Austria *July 2026*
- Third Derivative-Free Optimization Symposium, Université de Montréal *June 2026*
- SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom *June 2026*
- INFORMS Optimization Society Conference, Atlanta, GA *March 2026*

PDCS: A Primal-Dual Large-Scale Conic Programming Solver with GPU Enhancements

- INFORMS Annual Meeting, Atlanta, GA *October 2025*
- cuOpt Team, NVIDIA, virtual *July 2025*
- Computational Robotics Group Seminar, Harvard University, Cambridge, MA *May 2025*

New Theory and New Practical Methods for Solving Large-Scale Linear and Conic Optimization

- INFORMS Annual Meeting (Dantzig Dissertation Competition Session), Atlanta, GA *October 2025*
- PhD Thesis Defense, MIT, Cambridge, MA *May 2025*

Theoretical Foundations and Practical Improvements for Large-Scale Linear Programming

- Data Science Lab Seminar, MIT, Cambridge, MA *April 2025*
- Georgia Institute of Technology, Atlanta, GA *April 2025*
- Columbia University, New York, NY *February 2025*
- University of Chicago, Chicago, IL *February 2025*
- Cornell University, Ithaca, NY *January 2025*
- Northwestern University, Evanston, IL *January 2025*
- University of British Columbia, Vancouver, BC, Canada *January 2025*
- NC State University, Raleigh, NC *January 2025*
- Chinese University of Hong Kong (CUHK), Hong Kong *January 2025*
- Hong Kong University of Science and Technology (HKUST), Hong Kong *December 2024*

Level-Set Geometry and Improving the Performance of PDHG for Linear Optimization

- INFORMS Nicholson Student Paper Competition, Seattle, WA *October 2024*
- INFORMS Annual Meeting, Seattle, WA *October 2024*
- Cornell ORIE Young Researchers Workshop, Ithaca, NY *October 2024*
- OM Seminar, MIT, Cambridge, MA *September 2024*

Theoretical Complexity of PDHG for Linear Programs with Unique Optima

- INFORMS Annual Meeting (Optimization Society Award Session), Atlanta, GA *October 2025*
- ARC Colloquium, Georgia Tech, Atlanta, GA *September 2025*
- Modeling and Optimization: Theory and Applications (MOPTA), Lehigh University, PA *August 2024*
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada *July 2024*
- Conference in honor of Yinyu Ye's retirement, Shanghai, China *July 2024*

Improving the Geometry of (Conic) Linear Optimization Problems for the Primal-Dual Hybrid Gradient Method

- Modeling and Optimization: Theory and Applications (MOPTA), Lehigh University, PA *August 2024*
- INFORMS Optimization Society Conference, Houston, TX *March 2024*
- LIDS Conference, MIT, Cambridge, MA *February 2024*
- INFORMS Annual Meeting, Phoenix, AZ *October 2023*
- ORC Student Seminar, MIT, Cambridge, MA *October 2023*
- Shanghai University of Finance and Economics, Shanghai, China *September 2023*
- Workshop on Modern Continuous Optimization, Cambridge, MA *August 2023*

Geometric Condition Measures for the Primal-Dual Hybrid Gradient Method for Linear Programming

- SIAM Conference on Optimization (SIOPT), Seattle, WA *June 2023*
- ORC Student Seminar, MIT, Cambridge, MA *March 2023*

Using Taylor-Approximated Gradients to Improve the Frank-Wolfe Method for Empirical Risk Minimization

- INFORMS Annual Meeting, Indianapolis, IN *October 2022*
- ORC Student Seminar, MIT, Cambridge, MA *October 2022*
- International Conference on Continuous Optimization (ICCOPT), Lehigh University, Bethlehem, PA *July 2022*

From an Interior Point to A Corner Point: Smart Crossover

- INFORMS Annual Meeting, Indianapolis, IN *October 2022*

Interior-Point Methods Strike Back: Solving the Wasserstein Barycenter Problem

- INFORMS Annual Meeting, Seattle, WA *October 2019*
- "Heart of the Machine" Seminar, Shanghai, China *October 2019*

Computing the Wasserstein Barycenter Efficiently: A Structured Linear System and Customized Algorithms

- Seminar of New Advances in Theory and Application of ADMM, Shanghai, China *March 2019*

Service

Referee for

- Journals: Operations Research, Mathematics of Operations Research, SIAM Journal on Optimization, Mathematical Programming, INFORMS Journal on Computing, Journal of Optimization Theory and Applications, Tutorials in Operations Research, Optimization Letters, Journal of Machine Learning Research
- Conferences: AAAI, ICML, NeurIPS

Session (Co-)Chair for

- INFORMS Annual Meeting, San Francisco, CA *November 2026*
- Modeling and Optimization: Theory and Applications (MOPTA), Bethlehem, PA *August 2026*
- INFORMS Optimization Society Conference, Atlanta, GA *March 2026*
- INFORMS Annual Meeting, Seattle, WA *October 2024*
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada *July 2024*
- INFORMS Optimization Society Conference, Houston, TX *March 2024*
- INFORMS Annual Meeting, Indianapolis, IN *October 2022*

Other

Personal interests: Hiking, skiing, traveling, cooking Sichuan cuisine, and vibe-coding small apps for everyday use.